INTERNATIONAL SEARCH REPORT

International application No.
PCT/JP2022/030962

A. CLASSIFICATION OF SUBJECT MATTER

G02B 6/36(2006.01)i; ***G02B 6/02***(2006.01)i; ***G02B 6/024***(2006.01)i; ***G02B 6/04***(2006.01)i
FI: G02B6/36; G02B6/02 461; G02B6/024; G02B6/04 B

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G02B6/24; G02B6/255-6/27; G02B6/30-6/34; G02B6/36-6/43

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Published examined utility model applications of Japan 1922-1996
Published unexamined utility model applications of Japan 1971-2022
Registered utility model specifications of Japan 1996-2022
Published registered utility model applications of Japan 1994-2022

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	JP 2017-161836 A (OLYMPUS CORP) 14 September 2017 (2017-09-14) paragraphs [0013]-[0039], fig. 1-5, 9-11	1–6
Y		6–7
X	JP 2019-066771 A (SUMITOMO ELECTRIC INDUSTRIES) 25 April 2019 (2019-04-25) paragraphs [0015]-[0036], fig. 1-8	1-2, 4-5, 7
Y		6–7
A	US 2002/0081077 A1 (NAULT) 27 June 2002 (2002-06-27) fig. 5	1-7
A	WO 2019/044079 A1 (SUMITOMO ELECTRIC INDUSTRIES) 07 March 2019 (2019-03-07) paragraphs [0028]-[0053], fig. 1-6	1-7
A	JP 2008-165003 A (KYOCERA CORP) 17 July 2008 (2008-07-17) paragraph [0004]	1–7

☐ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
05 October 2022	**25 October 2022**
Name and mailing address of the ISA/JP **Japan Patent Office (ISA/JP)** **3-4-3 Kasumigaseki, Chiyoda-ku, Tokyo 100-8915** **Japan**	Authorized officer Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/JP2022/030962

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
JP 2017-161836 A	14 September 2017	US 2019/0004254 A1 paragraphs [0023]-[0049], fig. 1-5, 9-11	
JP 2019-066771 A	25 April 2019	US 2019/0101704 A1 paragraphs [0020]-[0045], fig. 1-8 EP 3474052 A1 CN 109613657 A	
US 2002/0081077 A1	27 June 2002	CN 1486442 A	
WO 2019/044079 A1	07 March 2019	US 2020/0166715 A1 paragraphs [0046]-[0072], fig. 1-6	
JP 2008-165003 A	17 July 2008	(Family: none)	